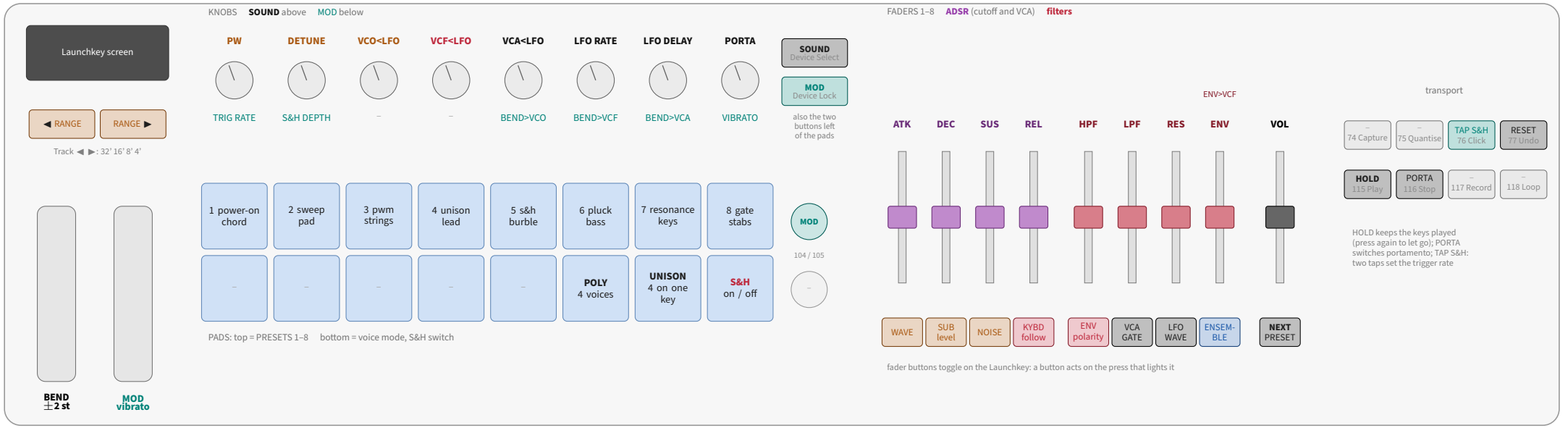


Signal path (as on the Jupiter-4's panel)

VCO (saw, square, pulse, PWM, off) + SUB (a square an octave down; off, low, mid, high) + NOISE → HPF → LPF → VCA → Ensemble → VOL, stereo out. One ADSR per voice moves the cutoff (ENV>VCF, either polarity) and the VCA; **VCA GATE** gives the VCA the key instead. One LFO (sine, square, saw, inverted saw) for all voices; the S&H samples noise at the trigger rate and steps the cutoff.

The filters

LPF: 4-pole (24 dB/oct), resonance limited as the Jupiter-4's diodes do; past **87 %** (fader 111) it **self-oscillates**, a sine at the cutoff. More resonance thins the bass, as on the original. KYBD follow: the cutoff follows the keys by 10, 40, 70 or 100 %; at 100 % a self-oscillating filter plays in tune. HPF: 1-pole, no resonance; off at the bottom.

Voices (bottom pads 6, 7)

POLY: four voices; a new key takes the voice released longest ago, or the oldest note. **UNISON**: all four on the lowest held key, spread by DETUNE (outer voices up to ±20 cents); a lower key takes over, releasing it slides back. PORTAMENTO glides each voice from its own last note. Each voice drifts a little in pitch (±3 cents), as analog ones do.

Oscillator (fader buttons 1–3, Track)

WAVE	saw → square → pulse → PWM → off
PW knob	pulse: 50, 40, 20, 10 % (four steps); PWM: the LFO's depth; saw, square: nothing
SUB	off → low → mid → high
RANGE	32', 16', 8', 4' (8' = the key's pitch)

Modulation

LFO to the pitch (VCO<LFO), the cutoff (VCF<LFO), the level (VCA<LFO, tremolo) and, on PWM, the pulse width. **LFO DELAY**: after a fresh key (all keys up before) the LFO fades in over that time. **S&H**: bottom pad 8 switches it; S&H DEPTH (MOD knob 2) sets how far the cutoff jumps, TRIG RATE (or TAP S&H) how often.

Wheels, Ensemble

BEND: the pitch ±BEND>VCO (2 st at power-on), and the cutoff and the level by BEND>VCF and BEND>VCA. MOD wheel: extra vibrato up to VIBRATO (0.47 st at power-on). **ENSEMBLE** (fader button 8): a stereo chorus, a slow 0.6 Hz swirl (rate and depth guessed, to tune by ear).

Inside

LPF: four trapezoidal (zero-delay) one-poles in a ladder, the resonance fed back through a soft clip; each sample solves one scalar equation, which has exactly one root, by 2 Newton steps (measured error $< 6 \cdot 10^{-5}$). It self-oscillates at the cutoff (within 0.1 %) from feedback 4. Oscillators at 48 kHz with 4-point BLEP edges (aliasing -57 dB at C4, -48 dB at C7). Envelopes per sample; pitch and cutoff every 4 samples. Code: `lib/quartet_voice.h`.

Load

DSP 8.5 % idle, 64.5 % with four voices held, 72 % with every source, full resonance and the Ensemble; patch RAM 19 of 44 KB.

Presets: top pads = 1–8; NEXT PRESET (fader button 9) steps; RESET (Undo) = 1

(2–8 set from the parameter meanings, not yet tuned by ear)

1 power-on chord	Saw through the LPF at 1.6 kHz, resonance 0.24, the envelope opens it 2.8 octaves; portamento 0.11 s. The power-on sound.
2 sweep pad	Saw + sub; a 1.5 s attack opens the cutoff 4 octaves from 240 Hz at resonance 0.6, a 0.3 Hz LFO moves it half an octave; long decay and release; Ensemble.
3 pwm strings	PWM at 0.7 Hz, 0.6 s attack, 1.2 s release, HPF at 86 Hz, LPF at 3.5 kHz; Ensemble.
4 unison lead	UNISON, outer voices 9 cents apart, portamento 0.2 s, vibrato 5.5 Hz fading in after 0.6 s; resonance 0.45.
5 s&h burble	Saw + sub, S&H on: 8 steps a second, up to 1.9 octaves, around a resonant 900 Hz.
6 pluck bass	16', saw + sub, no sustain: the envelope opens the cutoff 5 octaves from 230 Hz, resonance 0.6.
7 resonance keys	Sources off, resonance 1, keyboard follow 100 %: the filter sings the keys (C4 about 25 cents flat: tune with LPF).
8 gate stabs	VCA GATE, the envelope inverted on the cutoff (it dips and opens again), resonance 0.5.

A preset sets the switches, the voice mode, both knob pages and faders 1–8; the volume stays. After a preset a knob or fader moves its value from where it is (no jump); it catches up at the end of its travel at the latest.

Recipes

Filter as oscillator: WAVE off, SUB off, RES past 87 %, KYBD 100 %; tune with LPF. **Burble**: S&H on, RES at 3/4, S&H DEPTH at half, TRIG RATE to taste (or tap it). **Fat lead**: UNISON, DETUNE at 1/3, PORTA at 1/4. **Strings**: WAVE to PWM, PW at 3/4, LFO RATE at 1/4, ATK and REL at 3/4, ENSEMBLE. **Delayed vibrato**: VCO<LFO at 1/8, LFO RATE at 3/5 (5 Hz), LFO DELAY at 1/3 (1 s). **Bass**: RANGE 16', SUB high, SUS 0, DEC at 1/2, ENV>VCF at 2/3.

Where the values fall (knob or fader at . . .)

	0	1/4	1/2	3/4	full
ATTACK (s)	0.0006	0.005	0.044	0.35	3
DECAY, RELEASE (s)	0.014	0.073	0.38	1.9	10
HPF (Hz)	off	135	460	1500	5000
LPF (Hz, at C4)	20	114	650	3500	20000
RES (feedback)	0	1.2	2.3	3.4	4.6 (self-osc. from 4)
ENV>VCF (oct)	0	1.8	3.5	5.2	7
PW (pulse width)	50 %	40 %	20 %	20 %	10 %
PW on PWM (narrowest)	50 %	39 %	27 %	16 %	5 %
DETUNE (cents, outer)	0	5	10	15	20
VCO<LFO (st)	0	0.76	3	6.7	12
VCF<LFO (oct)	0	0.25	1	2.2	4
VCA<LFO (level dips by)	0	25 %	50 %	75 %	100 %
LFO RATE (Hz)	0.1	0.54	2.9	15	80
LFO DELAY, PORTA (s)	0	0.63, 0.13	2.5, 0.51	5.6, 1.1	10, 2
TRIG RATE (Hz)	1	2.3	5.1	11	25
S&H DEPTH (oct)	0	0.76	1.5	2.2	3
BEND>VCO / VCF / VCA	0	3 st / 0.5 oct / 3 dB	6 / 1 / 6	9 / 1.5 / 9	12 / 2 / 12
VIBRATO (st, wheel up)	0	0.25	0.5	0.75	1

Something seems wrong?

Silence: WAVE off with SUB and NOISE off (and RES below 87 %), LPF at the bottom, HPF at the top, SUS 0 with a short decay, or VOL down. **A button needs two presses**: the fader buttons toggle; they act on the press that lights them. **A knob does nothing**: PW acts only on pulse and PWM, DETUNE only in UNISON, S&H DEPTH only with S&H on; or the knob has not caught up yet: keep turning. **Notes that will not stop**: HOLD (Play). **Lost**: RESET (Undo). **Another patch** (on the SD card): hold both buttons left of the pads and tap a top pad: 1 freak, 2 grains, 3 groove, 4 soliton, 5 aeolian, 6 raw, 7 quartet.